



Data Warehouses

Lecture 4



Reminder

Last meeting:
OLAP in SQL
Execution plans

Lecture Goals

Goal:

Source data

Data quality

Data profiling

The "Unknown"

Definition: A null is not a zero, a blank space, or an empty string. It represents missing, inapplicable, or unknown data.

The OLAP Challenge: Analytical databases thrive on consistency. Nulls introduce ambiguity into otherwise deterministic data models.

Impact Areas: Nulls directly skew business intelligence by complicating two primary OLAP operations: Filtering and Aggregating.



Aggregates and NULL

The Danger of Averages and Counts

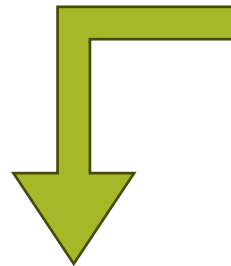
COUNT(*) vs. COUNT(Column):

- COUNT(*) counts rows (including those with nulls).

- COUNT(Sales) counts only non-null values in the Sales column.

Aggregates and NULL

```
SELECT
    count(manager_id),
    count(department_id),
    count(NULL),
    count(1),
    count(*),
    count(ROWNUM)
FROM departments;
```



	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	Administration	200	1700
2	20	Marketing	201	1800
3	30	Purchasing	114	1700
4	40	Human Resources	203	2400
5	50	Shipping	121	1500
6	60	IT	103	1400
7	70	Public Relations	204	2700
8	80	Sales	145	2500
9	90	Executive	100	1700
10	100	Finance	108	1700
11	110	Accounting	205	1700
12	120	Treasury	(null)	1700
13	130	Corporate Tax	(null)	1700
14	140	Control And Credit	(null)	1700
15	150	Shareholder Services	(null)	1700
16	160	Benefits	(null)	1700
17	170	Manufacturing	(null)	1700
18	180	Construction	(null)	1700
19	190	Contracting	(null)	1700
20	200	Operations	(null)	1700
21	210	IT Support	(null)	1700
22	220	NOC	(null)	1700
23	230	IT Helpdesk	(null)	1700
24	240	Government Sales	(null)	1700
25	250	Retail Sales	(null)	1700
26	260	Recruiting	(null)	1700
27	270	Payroll	(null)	1700

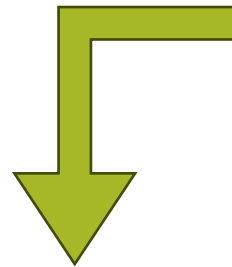
	COUNT(MANAGER_ID)	COUNT(DEPARTMENT_ID)	COUNT(NULL)	COUNT(1)	COUNT(*)	COUNT(ROWNUM)
1	11	27	0	27	27	27

Aggregates and NULL

SELECT

*sum(manager_id/manager_id),
sum(department_id/department_id),
sum(NULL),
sum(1)*

FROM departments;



DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10 Administration	200	1700
2	20 Marketing	201	1800
3	30 Purchasing	114	1700
4	40 Human Resources	203	2400
5	50 Shipping	121	1500
6	60 IT	103	1400
7	70 Public Relations	204	2700
8	80 Sales	145	2500
9	90 Executive	100	1700
10	100 Finance	108	1700
11	110 Accounting	205	1700
12	120 Treasury	(null)	1700
13	130 Corporate Tax	(null)	1700
14	140 Control And Credit	(null)	1700
15	150 Shareholder Services	(null)	1700
16	160 Benefits	(null)	1700
17	170 Manufacturing	(null)	1700
18	180 Construction	(null)	1700
19	190 Contracting	(null)	1700
20	200 Operations	(null)	1700
21	210 IT Support	(null)	1700
22	220 NOC	(null)	1700
23	230 IT Helpdesk	(null)	1700
24	240 Government Sales	(null)	1700
25	250 Retail Sales	(null)	1700
26	260 Recruiting	(null)	1700
27	270 Payroll	(null)	1700

	SUM(MANAGER_ID/MANAGER_ID)	SUM(DEPARTMENT_ID/DEPARTMENT_ID)	SUM(NULL)	SUM(1)
1	11	27	(null)	27



Aggregates and NULL

The Danger of Averages and Counts

(AVG):

AVG() typically ignores nulls entirely.

Example: Averaging [10, 20, Null, 30] divides the sum (60) by 3, yielding 20.

If the Null meant "0 sales," the true average should be $60 / 4 = 15$.

Business Impact: Inflated performance metrics and misleading KPIs.

Aggregates and NULL

Different COUNT behaviour:

`COUNT(*)`:

Counts the total number of rows in the group, regardless of whether any columns contain NULL values. It simply counts the rows.

`COUNT(expression)` or `COUNT(column_name)`:

Counts the number of rows in which the specified expression or column_name is not NULL. It ignores all rows in which the value is NULL.

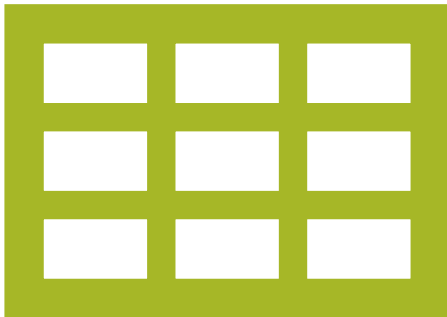
`COUNT(DISTINCT column_name)`:

Counts the number of unique values (excluding NULLs) in a specified column.

The `SUM()` function calculates the sum, `AVG()` calculates the average, and `MIN()`[`MAX()`] finds the minimum[maximum] of all non-NULL values in the group.

NULL values are completely ignored and do not affect the result.

Grouping and Dimensional Slicing



The "Null" Dimension:

When grouping data by a dimension containing nulls (e.g., GROUP BY CustomerSegment), the OLAP engine rolls all nulls into a single "Null" or "(Blank)" category.

The Problem:

This lumps entirely unrelated records together simply because they lack data in that specific column.

Drill-Down Issues:

End-users clicking into a "(Blank)" wedge on a pie chart are presented with a confusing, disjointed subset of fact rows.

Taming NULLS

Impute at the ETL Layer:

- Avoid bringing nulls into the OLAP cube whenever possible.

Use Default Values:

- Replace nulls with explicit "Unknown", "N/A", or "Not Categorized" string values for text dimensions.

Mathematical Safety:

- Replace nulls with 0 in numeric fact columns only if zero is the true business meaning of the missing data.

SQL Functions:

- Utilize COALESCE(column, default_value) or IFNULL() in metric definitions to handle nulls safely on the fly.



Source data

Data Sources

Data is diverse

Data sources:

Internal

External

Data structure:

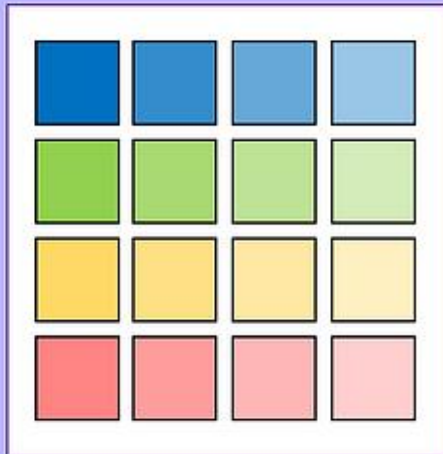
Structured

Semi-structured

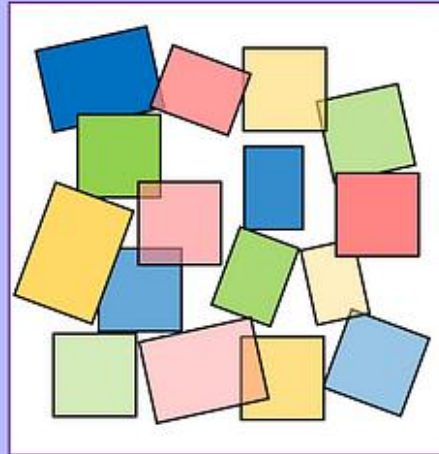
Unstructured



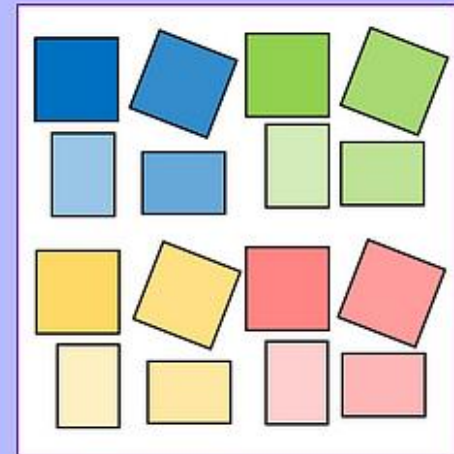
Data Sources



Structured Data
(Relational Databases, Tables)



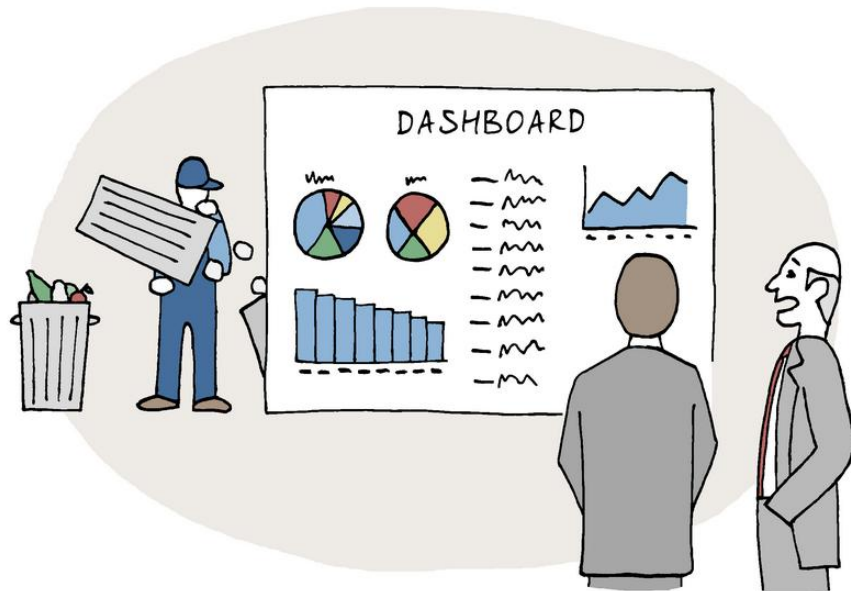
Un-structured Data
(Random Emails, Social Media Posts, Text Messages, Videos, Photos, Reviews)



Semi-structured Data
(JSON, XML, Files Labeled and Organized to Folder structure, Emails organized into mail boxes)



Data Quality



DO WE TRUST THIS DATA?

Data Quality

Garbage In,
Garbage Out

Data Quality



Data supports reasoning.

More data and analysis do not make better decisions.



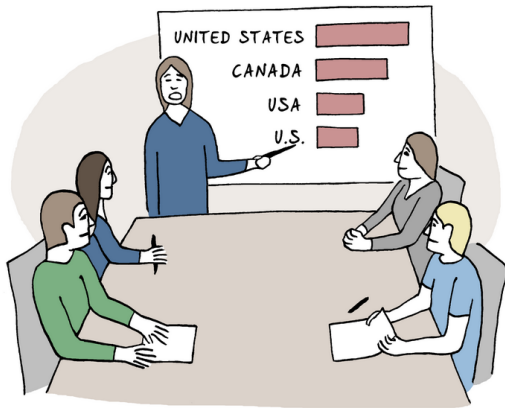
Facts supports decisions.

Facts = true, consistent, correct, clear information



Companies should rely on verified data that drives to commitment.

Data Quality



AS YOU CAN SEE, OUR TOP MARKETS ARE
UNITED STATES, CANADA, USA AND THE U.S.

 Dataedo /cartoon

Ploty@Dataedo

Many a BI project has been blindsided by data and data-quality issues.

- Often these issues are not apparent until users start testing the BI solution just before going “live.”

Happens often when people fail to assess the current state of their organization’s data

- get enamoured with the flashy BI products such as dashboards and data visualizations.

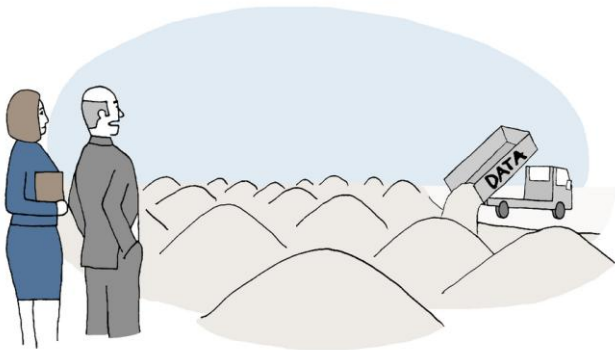
Data Quality

Most of that money has been spent in infrastructure,

whereas very little has been invested in managing and exploiting a valuable corporate asset a company's data

cannot be solved solely by the application of technology

the technology must augment a more serious senior-level management commitment to exploiting discovered knowledge and having a way to measure the value of those activities

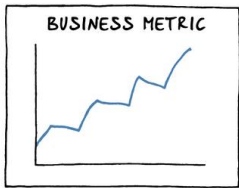


SO YOU'RE TELLING ME THIS WILL TURN INTO VALUE?

Data quality issues

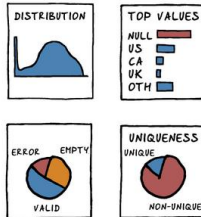
Typical data quality issues in Structured Data Sources (e.g., Relational Databases, Spreadsheets):

DATA ANALYSIS



„OUR BUSINESS IS BOOMING!“

DATA PROFILING



„OUR DATA IS C##P...“

Accuracy Issues

Completeness Issues

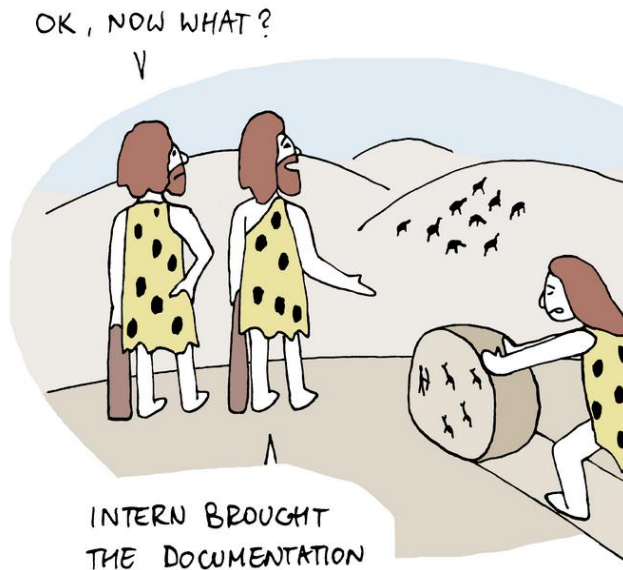
Consistency Issues

Uniqueness Issues (Duplication)

Validity Issues

Integrity Issues

Data quality issues



Issues in Semi-structured Data Sources
(e.g., JSON, XML, Logs):

- Structural Consistency Issues

 - (Schema Evolution/Variability)

- Data Type Consistency Issues

- Validity Issues

- Completeness Issues

- Accuracy & Uniqueness

- Complexity of nested structures

- Tag/key naming inconsistency

- Query performance

Data quality issues

Issues in Unstructured Data Sources (e.g., Text Documents, Emails, Images, Audio):

Ambiguity

Inconsistency

Lack of structure

Irrelevance (Noise)

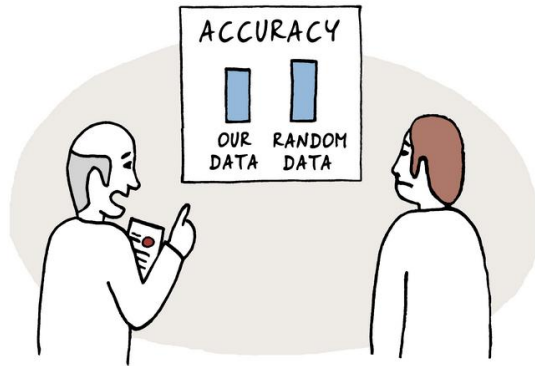
Extractability Issues

Accuracy (of Extracted Data)

Completeness (of Extracted Data)

Large Volume

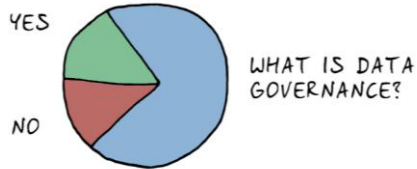
Lack of metadata and context



FUNNY STORY. I RAN SOME TESTS AND IT TURNS OUT THAT **RANDOM DATA** IS **MORE ACCURATE** THAN OUR DATA...

Data quality issues

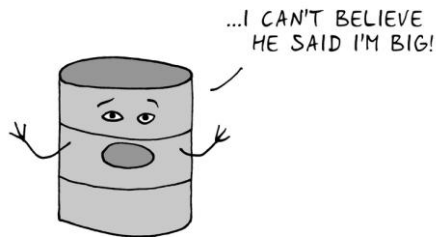
IS YOUR ORGANIZATION DOING
DATA GOVERNANCE?



Dataedo /cartoon

Piotr@Dataedo

SENSITIVE DATA



Dataedo /cartoon

Piotr@Dataedo

Other issues:

Data integration:

Combining data from different sources and with different structures is one of the biggest challenges in BI.

Metadata management:

The lack of a good description of data (its origin, meaning, transformation) makes it difficult to understand and use.

Data Governance:

The lack of clearly defined policies, roles, and responsibilities for data quality leads to chaos and low data reliability.

Security and privacy:

Ensuring compliance with regulations (e.g. GDPR) and protecting sensitive data is crucial.

Data Quality

WHERE IS SALES DATA?

INVOICES-TAB	CUSTOMER_TRANSACTIONS
TRX-CUST-HEADER-R	ORDERS-TAB
DM-ORDERS	FACTORDERS
DM-INVOICES	GL-TRANSACTIONS
FACTREVENUE	BILLING-DOCS
FACTTRANSACTIONS	INVOICE-LINES-TAB
TRX-CUST-LINES-R	ACCOUNT-TRANSACTIONS
TRANSACTIONS-SNAPSHOT	AR-TRANSACTION-HEADERS
SALES-TRANSACTIONS	AR-TRANSACTION-LINES

Data quality issues (operational databases integration problems)

No common time basis

Different calculation algorithms

Different levels of extraction

Different levels of granularity

Different data field names

Different data field meanings

Missing information

No data correction and validation rules

Different keys

Different language versions

Different data formats

Data Quality

All these source data quality issues create significant problems for data warehouses:

- ETL/ELT Failures

- Complex Transformation Logic

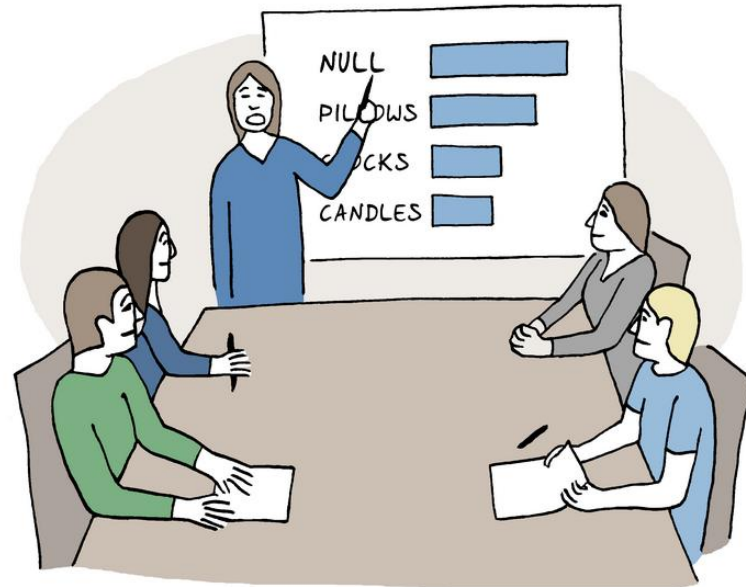
- Integration Challenges

- Inaccurate Reporting & Analytics

- Lack of User Trust

- Increased Maintenance

Data Quality



AS YOU CAN SEE, OUR BEST SELLING PRODUCT CATEGORY LAST YEAR WAS "NULL".

Data Quality

A quick search on subreddits for data engineers, data analysts, data scientists, and more can yield a plethora of users seeking data quality advice. And while the comment below may seem like the accepted way of doing data quality management...



SpookyScaryFrouze · 9mo ago
Senior Data Engineer

At my company, we usually wait for the stakeholders to send us a Slack message saying that the data seems wrong. We then spend a few hours investigating. Sometimes the data is wrong, sometimes not.

That's not how you're supposed to do it ?



220



Award



Share



... there's actually a much better way.

ONE YEAR LATER...



CREATE TABLE.
NAME... DOESN'T MATTER.
DONE.



WHAT THE #@%!
IS THIS TABLE?!

 Dataedo /cartoon

Piotr@Dataedo

Data Quality



REMEMBER KIDS,
DATA NEEDS TO BE MANAGED,
JUST LIKE ANY OTHER ASSET.

 Dataedo /cartoon

Piotr@Dataedo

Strategies for Working with Source Data

Early engagement

Thorough analysis of source data

Setting data quality standards

Implementation of data cleansing
processes

Data quality monitoring

Cooperation with source system owners

Data Quality

The key data attributes need to be assessed – 5C's:

Clean — is the data error free?

Consistent — are there many overlapping sources for data with inconsistent data?

Conformed — can the business analyse it across common, sharable dimensions?

Current — Is the data updated and available in the frequency needed?

Comprehensive — is the data needed for analytics even available or collected at this time?

You need to review the current state of the data

talk to subject matter experts on the source systems and perform a high-level data profiling exercise.

A critical assessment of the 5 C's will provide a data-quality baseline

determine the scope of the data integration and cleansing efforts that the BI project will need to provide the analytics that the business is requesting.

Data Quality

Clean

dirty data can really muddy up a company's attempt on data analysis, e.g., financial reports and based on incorrect information

Dirty data has missing items, invalid entries, and other problems that wreak havoc with automated data integration and data analysis.

Most source data is dirty to some degree, which is why data profiling and cleansing are critical steps in data warehousing.

Consistent

there should be no arguments about whose version of the data is the correct one.

Management meetings should never have to break down into arguments about whose number is correct when they really need to focus on how to improve customer satisfaction, increase sales, or improve profits.

Business people using different hierarchies or calculations for metrics will argue regardless of how clean the transactional data is.

Data Quality

Conformed

the business needs to analyse the data across common, shareable dimensions

business people across the enterprise are to use the same information for their decision-making

Current

the business needs to base decisions on whatever currency is necessary for that type of decision.

In some cases, such as detecting credit card fraud, the data needs to be up to the minute.

Comprehensive

business people should have all the data they need to do their jobs

regardless of where the data came from and its level of granularity

Data Quality

These dimensions are used to categorize data quality metrics:

Completeness:

This represents the amount of data that is usable or complete.

Uniqueness:

This accounts for the amount of duplicate data in a dataset.

Validity:

This dimension measures how much data matches the required format for any business rules.

Timeliness:

This dimension refers to the readiness of the data within an expected time frame.

Accuracy:

This dimension refers to the correctness of the data values based on the agreed upon “source of truth.”

Consistency:

This dimension evaluates data records from two different datasets.

Fitness for purpose:

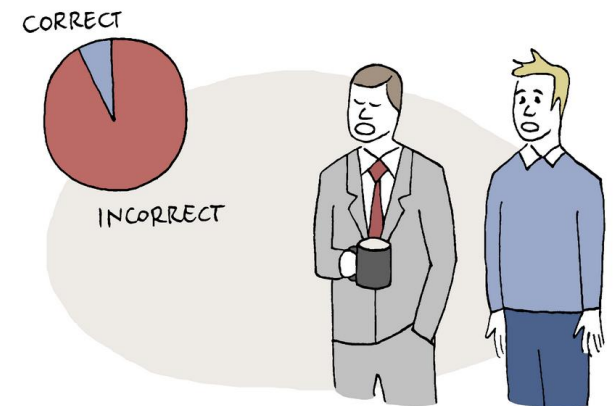
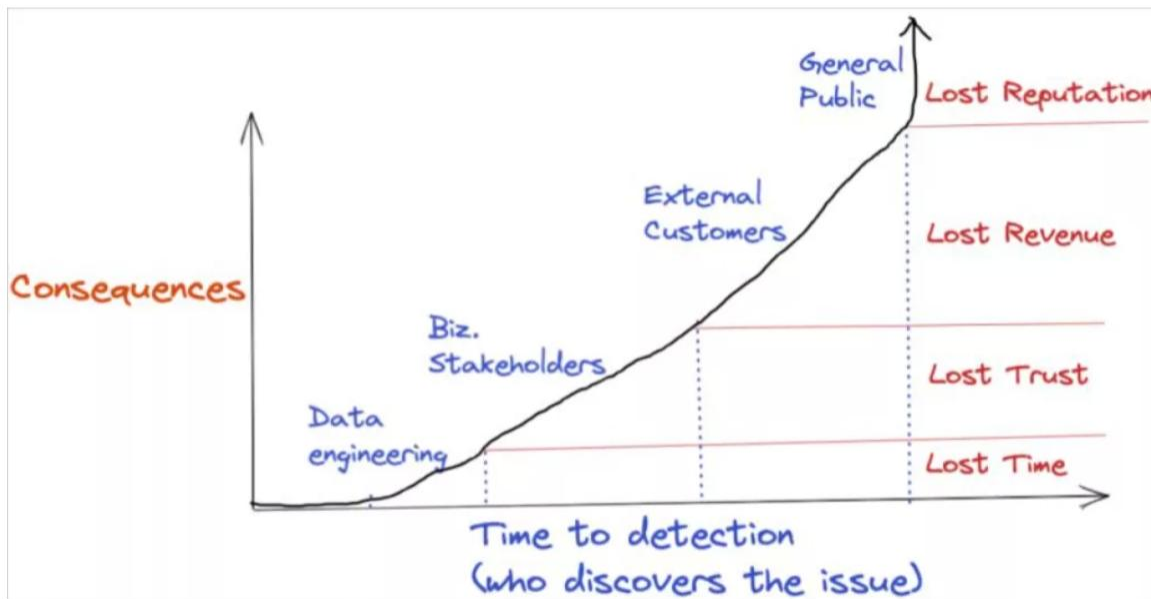
Evaluate how informative and useful data is for a given purpose. Helps to ensure that the data asset meets a business need.

5C and 6D

Kimball's 5 C's	Mapping to Technical Dimensions	Practical Engineering Check
Clean	Accuracy & Validity	Does the data match the real world? Does it follow format rules?
Consistent	Consistency	Do "Total Sales" match between the Finance and Marketing dashboards?
Conformed	Uniqueness & Consistency	Are we using a single, unique Master ID for each customer across all systems?
Current	Timeliness	Is the data fresh enough for the business requirement (Real-time vs. Daily)?
Comprehensive	Completeness	Are there missing records or null values in critical columns?

Kimball's business-oriented goals (The 5 C's) mapping to the technical DAMA dimensions (The 6 Dimensions).

Data Quality



KNOWING THE EXACT QUALITY OF OUR
DATA GIVES ME CONFIDENCE.



118%
OF ALL
DATA ANALYSTS
REPORT THEY HAVE
NEVER MADE AN
ERROR
IN THEIR REPORT

 Dataedo /cartoon Prot@Dataedo

Data Profiling

learn about the data itself - its fitness, quality, scope, etc.

Data profiling



Data profiling

Data profiling is the process of reviewing source data, understanding structure, content and interrelationships, and identifying potential for data projects.

panoply

Data profiling, the act of monitoring and cleansing data, is an important tool organizations can use to make better data decisions

SAS

Data profiling is the process of examining, analyzing, and creating useful summaries of data. The process yields a high-level overview which aids in the discovery of data quality issues, risks, and overall trends.

Talend



Data profiling, or data archeology, is the process of reviewing and cleansing data to better understand how it's structured and maintain data quality standards within an organization. The main purpose is to gain insight into the quality of the data by using methods to review and summarize it, and then evaluating its condition.

IBM

Data profiling is the systematic up-front analysis of the content of a data source, all the way from counting the bytes and checking cardinalities up to the most thoughtful diagnosis of whether the data can meet the high-level goals of the data warehouse.

Kimball Ralph

Users' understanding of Data

improve data quality, shorten the implementation cycle of major projects, and improve users' understanding of data



Data Content and Structure

understanding structure, type, content and interrelationships, and identifying potential for data projects.

Data Discovery

Discovering business knowledge embedded in data



Data Profiling Benefits



Find Data Problems

Find data related problems in advance and thus helps in remediating those issues

Sensitive Data

Helps in identifying gaps between sensitive data



Data Insights and Opportunities

Helps in accurately analysing data which can lead to new insights and opportunities as we treat Data as corporate asset

Data profiling

Crucial part of:

Data warehouse and business intelligence projects

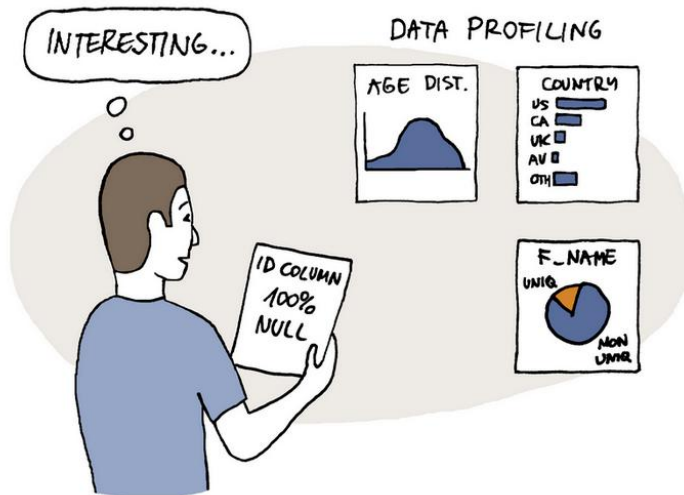
uncover data quality issues in data sources – what needs to be corrected in ETL.

Data conversion and migration projects

identify data quality issues and uncover new requirements for the target system

Source system data quality projects

highlight data which suffers from serious or numerous quality issues, and the source of the issues



Data profiling involves:

Structure

Table and column

Collecting data types, length

Identifying key candidates, foreign-key candidates, functional dependencies, performing inter-table analysis.

Quality

Performing data quality assessment – completeness, accuracy, consistency, timeliness

Discovering metadata and assessing its accuracy.

Business rules

Content

Collecting descriptive statistics like min, max, count and sum.

Recurring patterns, Identifying distributions/frequencies

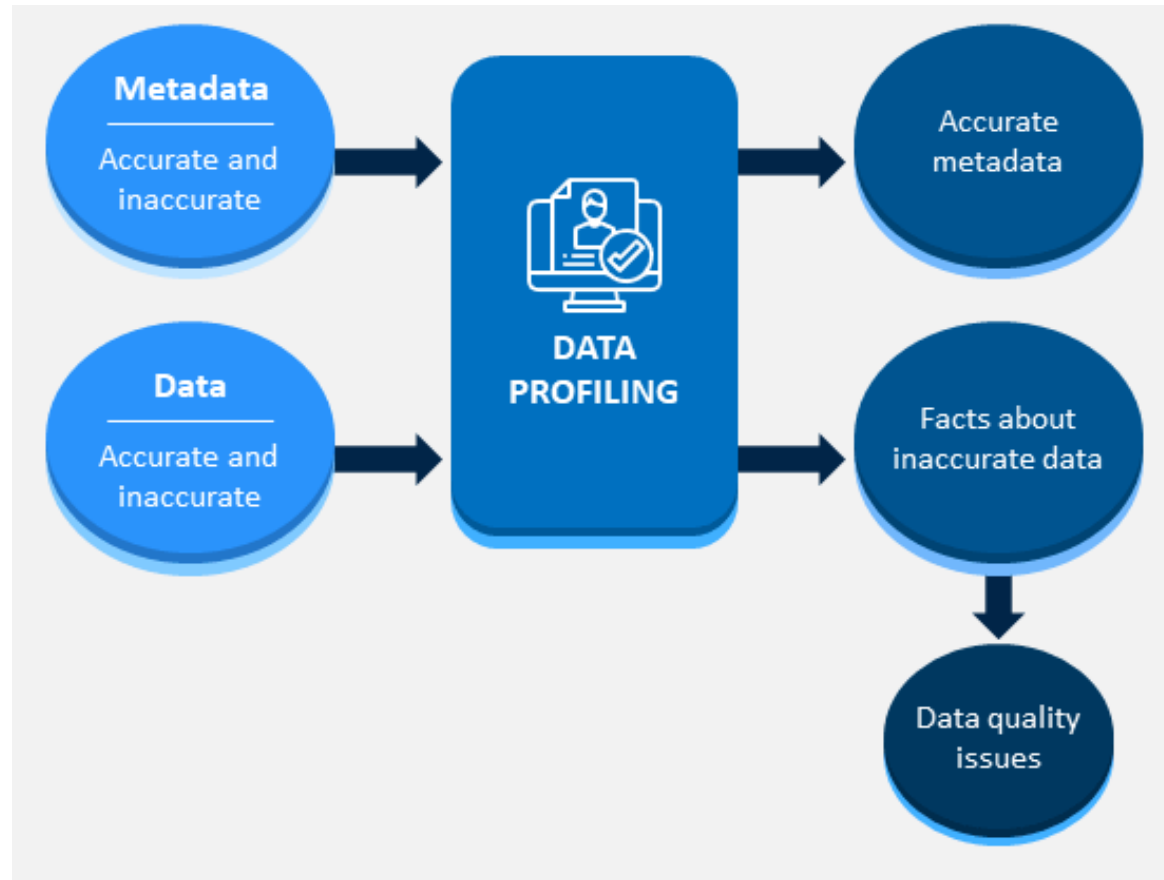
Relationships between data

Tagging data with keywords, descriptions or categories.

Data profiling

Data profiling

Data profiling enables you to discover many important things about your data.



Data profiling

Some common findings include the following:

- A domain of valid product codes

- A range of product discounts

- Columns that hold the pattern of an e-mail address

- A one-to-many relationship between columns

- Anomalies and outliers within columns

- Relations between tables even if they are not documented in the database

Data profiling

Data profiling isn't a monolithic activity.

The specific techniques we employ and the insights we seek depend heavily on the characteristics of the data attributes themselves.

We need to consider:

Technical Data Type: (e.g., String, Integer, Date, Boolean) - dictates basic operations.

Usage or Role: (e.g., Is it an Identifier, a Descriptive attribute, a Measure?) - dictates its purpose.

Measurement Scale: (Nominal, Ordinal, Interval, Ratio) - dictates appropriate statistical analysis.

Understanding these characteristics allows us to apply the most relevant profiling methods.

"Our profiling approach for each column must be tailored based on its attributes"

Data profiling

Attribute types by data type:

Text

They store data in the form of text. Enable alphabetical comparisons and grouping.

Analyze length distribution (min, max, average), cardinality (number of distinct values), frequency distribution of top/bottom values, pattern analysis (using regular expressions to check format)

Numeric

They store numeric values that can be used in calculations and groupings.

Calculate full statistical profile: min, max, mean, median, mode, standard deviation, variance, quartiles. Look for outliers

Boolean/Binary

They only take two values: True/False.

Check the distribution of True/False values, count nulls

Date/Datetime

They store date information in a specific format. They can be used for temporal comparisons.

Check for format consistency, logical range (min/max dates), identify gaps or unexpected values, analyze recency

Categorical

Focus on cardinality, frequency distribution, and validating against a known set of allowed values (domain check)

Attribute types by data use:

Identification attributes (keys)

Used to uniquely identify records: natural keys and surrogate keys

Critical Check: Uniqueness (cardinality should equal record count), null count (should be zero for PKs), format consistency

Foreign keys

Check referential integrity (do all FK values exist in the referenced Primary Key column?), nullability (depending on business rules)

Descriptive attributes

They store information that characterizes the object (e.g. customer name, product color, address).

Often used in filtering and grouping data.

Focus on understanding the values, their distribution, and identifying potential quality issues like typos or inconsistencies. Cardinality is important

Measures/facts

Numeric values to be rolled up (e.g., 'revenue', 'quantity').

Focus on statistical properties (mean, sum, range, outliers)

Flags/statuses

They store Boolean or categorical values (e.g., 'order_status').

Focus on cardinality, frequency distribution, and validating against a known set of allowed values (domain check), business coherence

Calculated fields

Generated from other columns (e.g. 'profit = revenue - cost').

Focus on business rules coherence.

Data profiling

Attribute types by data nature:

Nominal

Eye color (brown, blue), gender (male, female), product type (TV, refrigerator).

Ordinal

The level of customer satisfaction ("very satisfied", "satisfied", "neutral").

Interval

Temperature in Celsius, date (e.g. the difference between dates).

Ratio

The weight of the product, the customer's income, the number of units sold.

Misinterpreting the measurement scale leads to incorrect analysis

e.g., calculating the average 'PostalCode' (Nominal) is nonsensical.

THE FOUR LEVELS OF MEASUREMENT:

	Nominal	Ordinal	Interval	Ratio
Categorizes and labels variables	✓	✓	✓	✓
Ranks categories in order		✓	✓	✓
Has known, equal intervals			✓	✓
Has a true or meaningful zero				✓

Data Profiling

Types of data profiling

Structure discovery – helps understand how well data is structured.

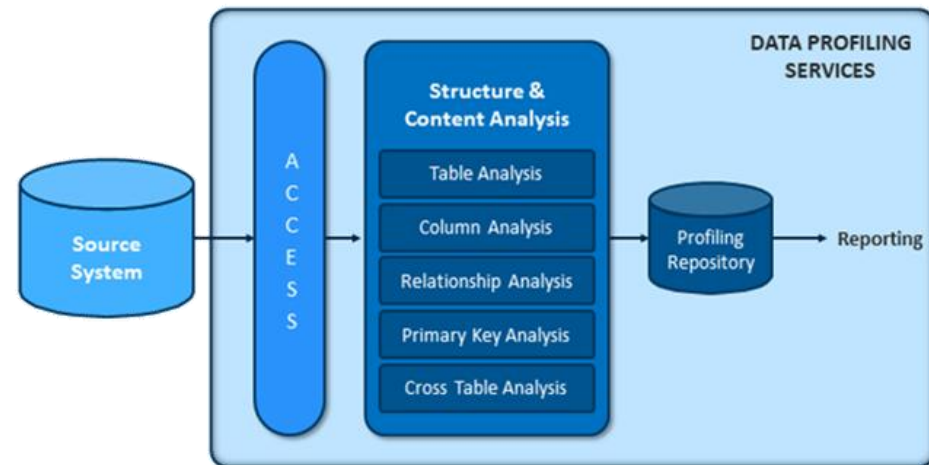
for example, what percentage of phone numbers do not have the correct number of digits.

Content discovery – looking into individual data records to discover errors.

for example, phone numbers with no area code.

Relationship discovery – how parts of the data are interrelated.

for example, key relationships between database tables, references between cells or tables in a spreadsheet.



Key Aspects of Data Profiling

Data Structure Analysis:

Data Types: Determining the data type of each attribute (e.g., integer, string, date).

Data Length: Analyzing the length and size of data fields.

Data Distribution: Examining how data is distributed across different values (e.g., histograms, frequency tables).

Data Relationships Analysis:

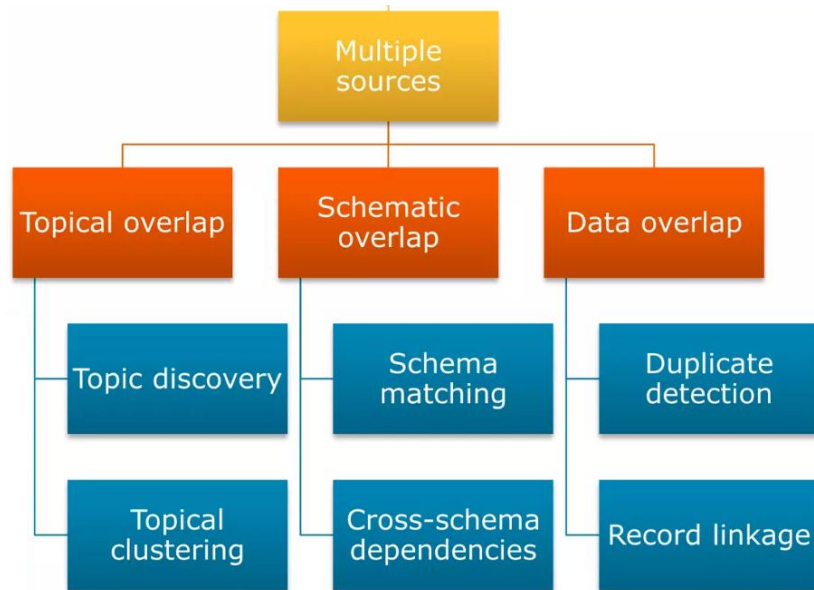
Identifying Relationships: Discovering relationships between different data attributes.

Dependency Analysis: Determining dependencies between attributes.

Metadata Collection:

Gathering metadata about the data, such as data sources, data lineage, and data quality rules.

Data Profiling



Multiple sources

Additional level of complexity



Data profiling



Data profiling sifts through data to determine its legitimacy and quality

Beyond just ensuring that your data is up to standard statistical measures, data profiling should also focus on business rules specific to a company.

Data Profiling

Data profiling steps (R. Kimball):

Discover if data is suitable for analysis

a “go / no go” decision on the project

Identify and correct data quality issues in source data

before starting to move it into target database

Identify data quality issues that can be corrected during Extract-Transform-Load (ETL)

can uncover if additional manual processing is needed.

Identify unanticipated business rules, hierarchical structures and foreign key / private key relationships

use them to fine-tune the ETL process.

Series of tests starting with individual fields and ending with whole suites of tables

Individual fields are checked to see that their contents agree with their basic data definitions and domain declarations.

It is especially valuable to see how many rows have null values, or have contents that violate the domain definition.

For example, if the domain definition is “telephone number” then alphanumeric entries clearly represents a problem.

The best data profiling tools count, sort, and display the entries that violate data definitions and domain declarations

Beyond single fields – relation **between fields**

Describe the relationships discovered between fields in the same table.

Fields that implement a key to the data table, together with higher level many-to-1 relationships that implement hierarchies

Checking what should be the key of a table is especially helpful because the violations (duplicate instances of the key field) are either serious errors, or reflect a business rule that has not been incorporated into the ETL design.

Relationships between tables

including assumed foreign key to primary key relationships and the presence of parents without children.

Check complex **business rules** unique to a business

Such as verifying that all the preconditions for a customer status have been met.

Data profiling

Data Profiling

Basic tools:

Column profiling:

Scan tables and counts the number of times each value shows up within each column.

Can be useful in finding frequency distribution and patterns within a column.

Distinct count and percent – identifies natural keys, distinct values in each column that can help process inserts and updates.

Crucial for tables without headers.

Percent of zero / blank / null values – identifies missing or unknown data.

Helps ETL architects setup appropriate default values.

Minimum / maximum / average string length – supports selection of appropriate data types and sizes in target database.

Enables setting column widths just wide enough for the data, to improve performance.

Cross-column profiling:

Two processes: key analysis and dependency analysis.

The key analysis process looks at the array of attribute values by scouting for a possible primary key.

While the dependency analysis process works to identify what relationships or patterns are embedded within the data set.

Cross-table profiling:

Use key analysis to identify stray data.

The foreign key analysis identifies orphaned records or general differences to examine the relationship between column sets in different tables.

Key integrity - ensures keys are always present in the data, using zero/blank/null analysis. Also, helps identify orphan keys, which are problematic for ETL and future analysis.

Data rule validation:

Assess data sets against established rules and standards to verify that they're in fact following those predefined rules.

Key Integrity:

Ensuring keys are always present in the data and identifies orphan keys, which can be problematic.

Cardinality:

Check relationships such as one-to-one and one-to-many, between data sets.

Pattern and frequency distribution:

Ensure data fields are formatted correctly.

for example if emails are in a valid format.

Extremely important for data fields used for outbound communications (emails, phone numbers, addresses)

```

SELECT MIN(LEN(mycolumn)) AS string_length_min,
       MAX(LEN(mycolumn)) AS string_length_max,
       AVG(LEN(mycolumn)) AS string_length_avg,
       STDEV(LEN(mycolumn)) AS string_length_stddev,
       VAR(LEN(mycolumn)) AS string_length_var
FROM mytable
WHERE mycolumn IS NOT NULL
AND LEN(mycolumn) > 0

```

	string_length_min	string_length_max	string_length_avg	string_length_stddev	string_length_var
1	9	41	20	5.56730064350621	30.9948364551846

```

DECLARE
@min int = 10,
@max int = 200;

```

	below_range	withn_range	above_range
1	13	93	189

```

SELECT
    SUM(CASE WHEN mycolumn < @min THEN 1 ELSE 0 end) AS below_range,
    SUM(CASE WHEN mycolumn BETWEEN @min and @max THEN 1 ELSE 0 end) AS within_range,
    SUM(CASE WHEN mycolumn > @max THEN 1 ELSE 0 end) AS above_range
FROM mytable

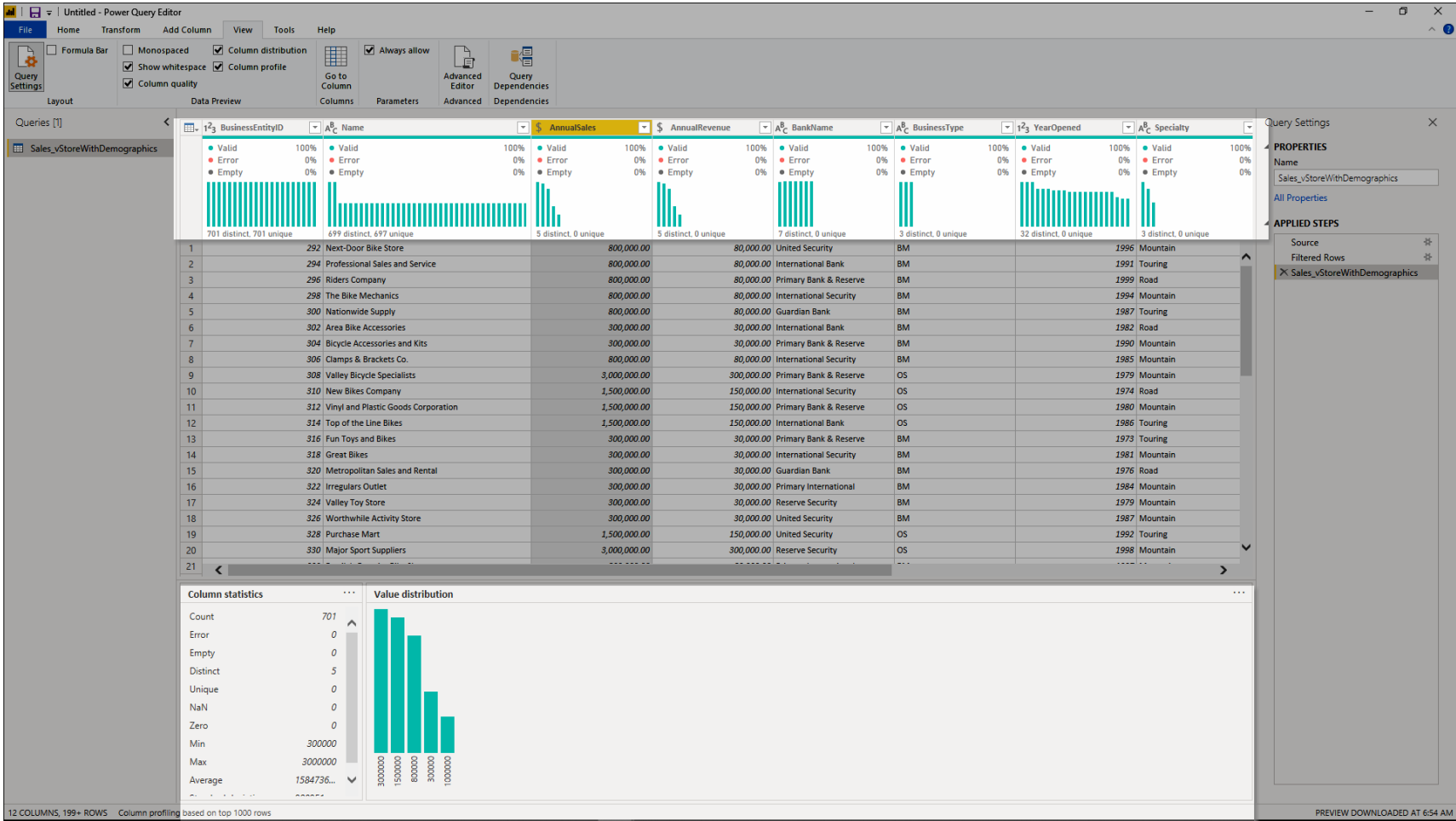
```

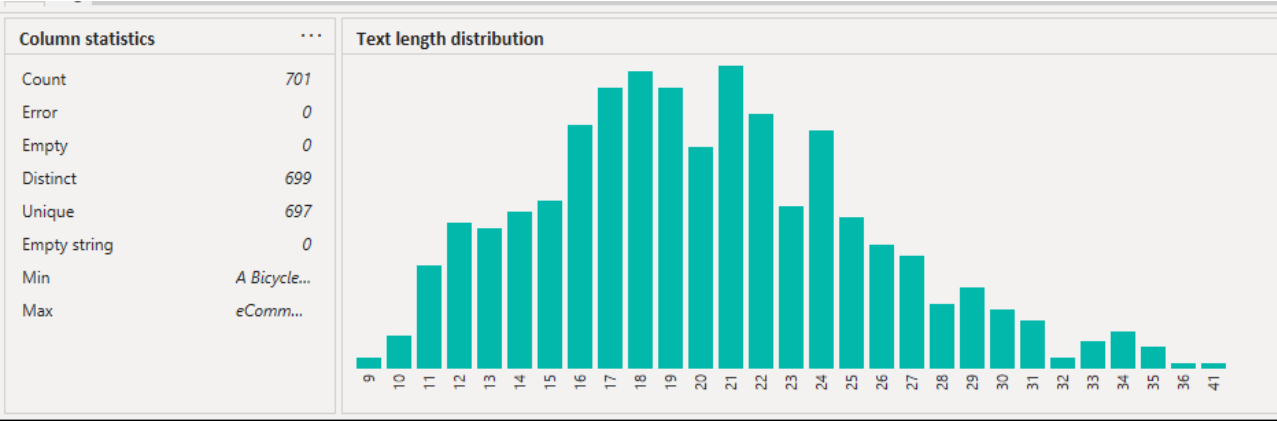
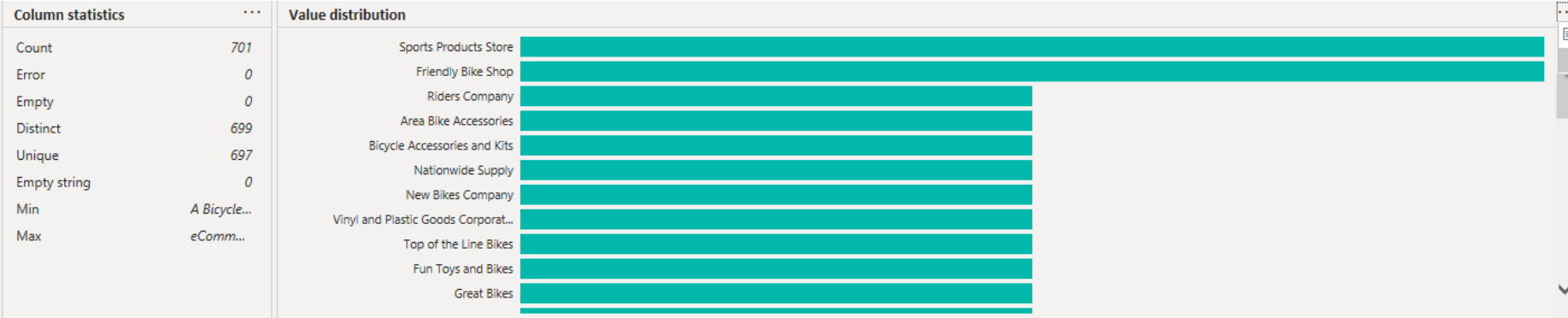
```

SELECT MIN(mycolumn) AS value_min_string,
       MAX(mycolumn) AS value_max_string
FROM mytable
WHERE mycolumn IS NOT NULL
AND LEN(mycolumn) > 0

```

	value_min_string	value_max_string
1	A Bike Store	Yellow Bicycle Company





Overview

Overview

Alerts 14

Dataset statistics

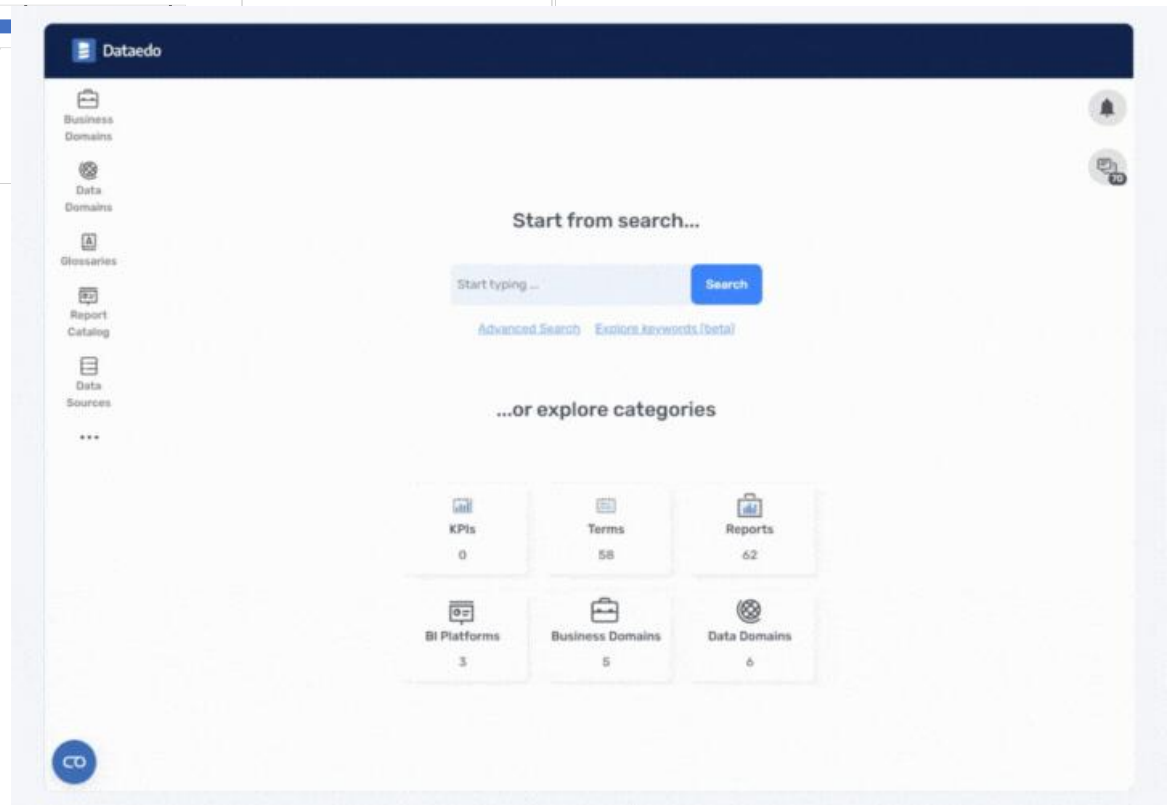
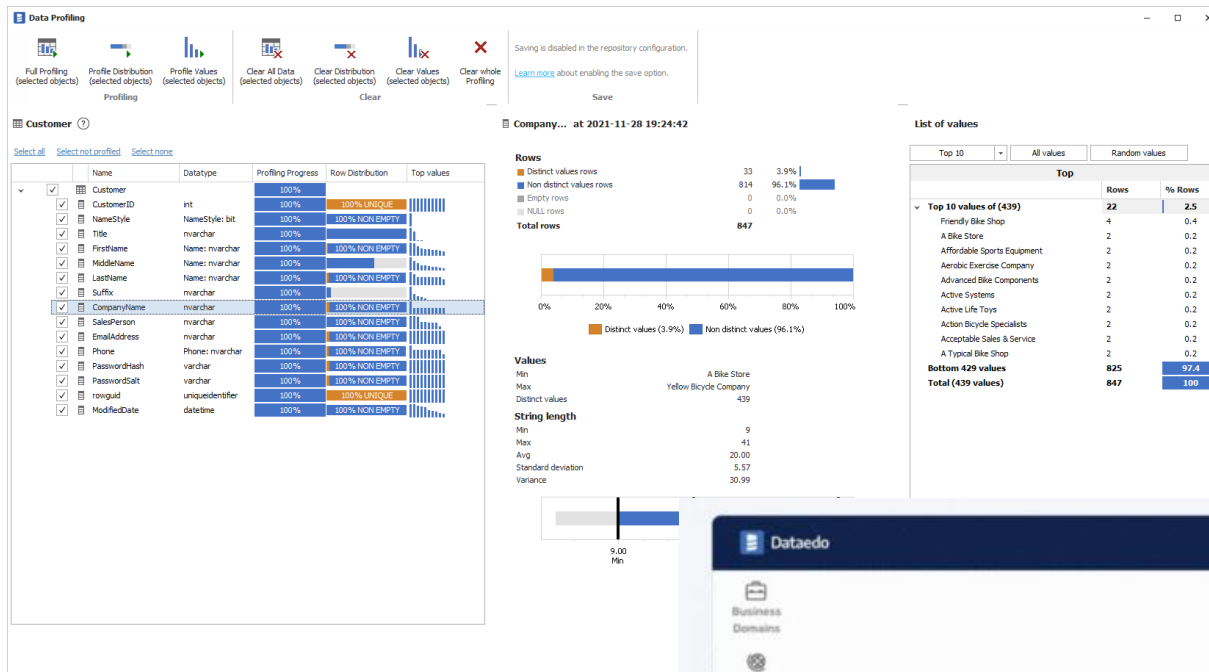
	Original	Synthetic
Number of variables	15	15
Number of observations	10000	70000
Missing cells	0	0
Missing cells (%)	0.0%	0.0%
Total size in memory	1.1 MiB	8.0 MiB
Average record size in memory	120.0 B	120.0 B

Variable types

	Original	Synthetic
Numeric	5	5
Categorical	7	7
Text	3	3

Variables

pandas Profiling (ydata-profiling)



<https://dataedo.com/kb/query/sql-server/data-profiling-data-quality>

Data Profiling



"I have come to the conclusion that data profiling should be the mandatory "next step" in every data warehouse project after the business requirements gathering."

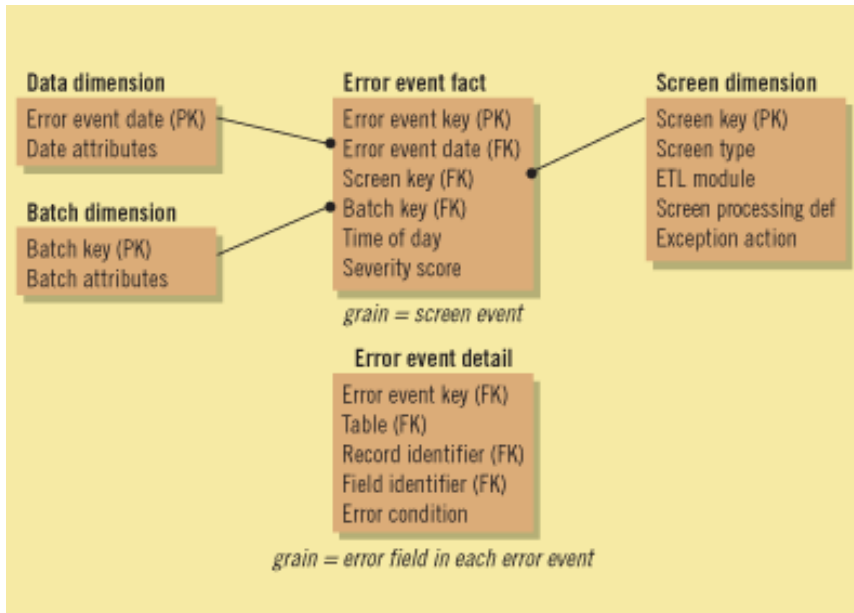
R. Kimball

"Big benefit of data profiling is that data profiling makes the implementation team look like they know what they are doing. "

R. Kimball

Correctly anticipating the difficult data quality issues of a project upfront, avoiding surprises of discovering BIG problems near the end of a project.

Quality screens



Quality Screens

a set of diagnostic filters
each implement a test in
the data flow

*if it fails it records an error in
the Error Event Schema*

Error Event Schema

place, where all error
events thrown by quality
screens, are recorded
maintained in multi-
dimensional structure

State	Count(*)
Rhode Island	1
Mississippi	2
New Yourk	5
Connecticut	7
New Mexico	43,844
Vermont	64,547
Mississippi	78,198
Utah	128,956
Wyoming	137,630
Missouri	148,953
Rhode Island	182,067
Minnesota	195,197
North Dakota	203,286
Michigan	241,245
Washington	274,528
Pennsylvania	287,289
Montana	337,128
Louisiana	341,972
Virginia	346,691
West Virginia	359,848
Delaware	422,091
Iowa	456,302
Massachusetts	480,958
Tennessee	483,688
New York	494,332
Nevada	506,235
South Dakota	514,485
Indiana	553,267
Connecticut	562,436

Quality screens

Example – Data Sampling

The simplest way to check for basic anomalies is to count the rows in a table while grouping on the column in question

This simple query reveals the distribution of values and displays potentially corrupt data

```
select state, count(*)
from order_detail
group by state
```

Quality Screen

The Challenge:

Identify the error and name the specific "Quality Screen" or "Profiling Technique" that would catch it.

Sale_ID	Trans_Date	Cust_ID	Amount	Category
1001	2023-10-15	C-99	150.00	Electronics
NULL	2023-10-16	C-10	45.00	Home
1003	2029-01-01	C-12	200.00	Toys
1004	2023-10-16	X-999	-10.00	[Unknown]
1005	2023-10-17	C-15	30.00	Garden

Quality screens

Quality screens are divided into three categories:

Column screens.

Used to test individual column,

e.g. for unexpected values like NULL values; non-numeric values that should be numeric; out of range values; etc.

Structure screens.

Used to test for the integrity of different relationships between columns (typically foreign/primary keys) in the same or different tables.

Used for testing that a group of columns is valid according to some structural definition it should adhere.

Business rule screens.

Used to test whether data, maybe across multiple tables, follows specific business rules

e.g. if a customer is marked as a certain type of customer, the business rules that define this kind of customer should be adhered

Quality screens

Known Table Row Counts

In some cases the number of records to be expected of a given data type from a given data provider is known.

The known table record count case can be handled by simple screen SQL, such as the following:

```
SELECT COUNT(*)  
FROM work_in_queue_table  
WHERE source_system_name =  
'Source System Name'  
HAVING COUNT(*) <>  
'Known_Correct_Count'
```

Column Length Restriction

Screening on the length of strings in textual columns is useful in both staged and integrated record errors.

An example of a SQL SELECT that performs such a screening:

```
SELECT  
unique_identifier_of_offending_records  
FROM work_in_queue_table  
WHERE source_system_name =  
'Source System Name'  
AND LENGTH(numeric_column) IS NOT  
BETWEEN min AND max.
```

Quality screens

Column Numeric and Date Ranges

from a data-quality perspective data may have ranges of validity that are restrictive

Column Explicit Valid Values

A given column may have a set of known discrete valid values as defined by its source system

Column Explicit Invalid Values

A given column is routinely populated with values known to be incorrect and for which there is no known set of discrete valid values,

explicitly screen for these invalid values.

Checking Table Row Count Reasonability

This class of screens is quite powerful but a bit more complex to implement.

Checking Column Distribution Reasonability

The ability to detect when the distribution of data across a dimensional attribute has strayed from normalcy is another powerful screen.

"Discovery-to-Enforcement" Workflow

The core of the integration clarity lies in how information flows from one activity to the other:

Phase 1: Profiling (Analysis)

Action: Technical exploration of the source data

e.g., frequency distributions, null counts, pattern matching.

Output: Data Quality Insights

e.g., "The Email column is 15% null and contains 2% invalid formats".

Phase 2: Rule Definition (Translation)

Action: Collaborating with stakeholders to decide which "As-Is" issues are unacceptable.

Output: Data Quality Requirements

e.g., "The Email column must be 100% populated and follow standard regex".

Phase 3: Quality Screens (Enforcement)

Action: Hard-coding those requirements into the ETL/ELT pipeline.

Output: Automated Pass/Fail gates and an Error Event Schema.

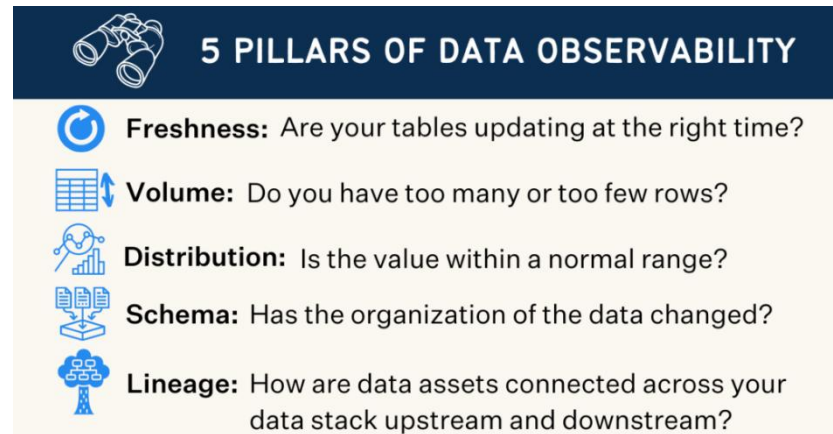
Data observability

Data observability

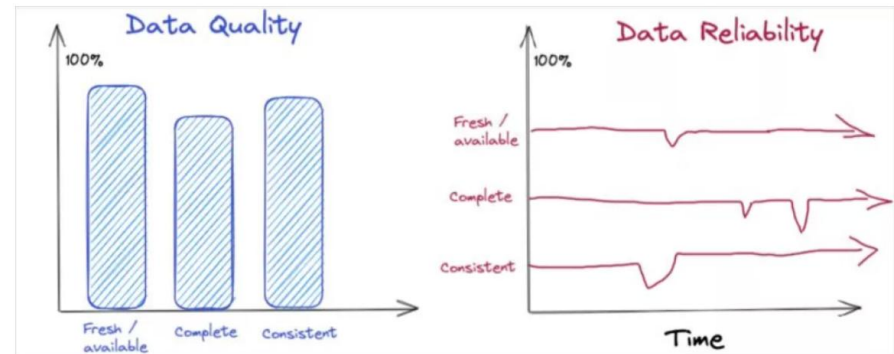
refers to an organization's comprehensive understanding of the health and performance of the data within their systems.

tools employ automated monitoring, root cause analysis, data lineage, and data health insights to proactively detect, resolve, and prevent data anomalies.

is the degree of visibility you have into your data at any point in time – it includes monitoring, troubleshooting, and ensuring data quality throughout the data stack.



Data quality vs data reliability



Data quality vs data reliability. Image courtesy of Shane Murray.

From Quality to Observability

Feature	Data Quality (Static)	Data Observability (Continuous)
Scope	Focuses on the Data itself (values).	Focuses on the System (pipelines, infrastructure, and data).
Detection	Reactive: Checks run after data is loaded (Unit Tests).	Proactive: Uses ML to detect "anomalies" in volume or arrival time.
State	Point-in-time: Is this table valid <i>right now</i> ?	Lineage-aware: Where did this break in the upstream flow?
Common Metric	"Does this column have nulls?"	"Why did the 8:00 AM load only have 10% of the usual volume?"

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